NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES (FAST-NUCES)

Chiniot-Faisalabad Campus — CL3001: Machine Learning Lab (Spring 2026)

Deliverable 1: Semester Project Paper Approval Form

Student Name:	Musa Rashid	**Instructor:**	Ms. Esha Iftikhar
Application Area:	Finance & FinTech (Credit Risk)	**GitHub Repository:**	github.com/musarashid49/credit-risk-calibration-ml

1. SELECTED RESEARCH PAPER (2026 PEER-REVIEWED JOURNAL)

- **Paper Title:** *Performance Evaluation of Machine Learning and Deep Learning Models for Credit Risk Prediction*
- **Authors:** Irvine Mapfumo and Thokozani Shongwe (University of Johannesburg)
- **Journal & Date:** *Journal of Risk and Financial Management* (MDPI), Vol. 19, Issue 3, Article 210 (**Published: March 2026**)
- **Paper Link (DOI):** https://doi.org/10.3390/jrfm19030210

2. PUBLICLY AVAILABLE DATASETS USED

- **Dataset 1 — Taiwanese Credit Card Default Dataset (UCI ML Repository):**
  https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients
  Contains **30,000 client records** and **24 financial features** (credit limit, demographics, 6 months of repayment status and bill statements). Imbalanced binary target: 77.88% Non-Default vs. 22.12% Default.
- **Dataset 2 — Statlog German Credit Dataset (UCI ML Repository):**
  https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data
  Contains **1,000 loan applicant records** and **20 financial/demographic features**. Imbalanced binary target: 70.0% Good Credit Risk vs. 30.0% Bad Credit Risk (Default).

3. ALIGNMENT WITH COURSE METHODOLOGY & REPORTED METRICS

- **Pre-Processing & Class Balancing:** Feature scaling, encoding, and hybrid resampling comparing **SMOTE**, **ENN** (Edited Nearest Neighbors), and **SMOTE-ENN**.
- **Classification Models Taught in Course:** Logistic Regression (LR), Decision Tree (DT), Support Vector Machine (SVM), Random Forest (RF), AdaBoost, XGBoost, and Multilayer Perceptron (MLP).
- **Evaluation Protocol & Baseline Metrics:** 10-fold stratified cross-validation reporting Accuracy, Precision, Recall, F1-Score, and ROC-AUC (reporting **95.4% Accuracy / 0.928 F1** with MLP + SMOTE-ENN on German Credit, and **82.1% Accuracy / 0.789 F1** with Random Forest + SMOTE-ENN on Taiwanese Credit).

4. REASON FOR SELECTION & PLANNED IMPROVEMENT (STEP 6 & STEP 7)

We selected this March 2026 study because automated credit risk underwriting is a vital real-world financial problem, its tabular datasets are fully reproducible on standard compute, and it presents a compelling scientific opportunity for **Step 6 (Critical Analysis)** and **Step 7 (Bonus Improvement)**:

- **Critical Replication Audit:** Hybrid resampling (SMOTE-ENN) applied prior to cross-validation splits causes boundary leakage, and altering class priors distorts predicted **Probability of Default (PD)** estimates required under Basel III banking standards.
- **Proposed Improvement:** We will implement a strict in-fold resampling pipeline paired with **Isotonic/Platt Probability Calibration** (evaluated via Brier Score and Expected Calibration Error) and **Instance-Dependent Cost-Sensitive Thresholding** using each borrower's actual credit limit (`LIMIT_BAL`) to minimize total financial portfolio loss.

Student Signature:	Instructor Approval:	Date:
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